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Paper Practice 9.1

Exponential Functions

Show all of your work in the space provided. Round money to the nearest dollar unless told otherwise.

1. Evaluate each function at the given value. (Warm-Up Review)

a. $f(x) = 4^x$, when $x = 0$

b. $f(x) = 3(2)^x$, when $x = 3$

c. $f(x) = 5(\frac{1}{2})^x$, when $x = 2$

2. Determine whether each function shows exponential growth or decay. (Growth or Decay)

a. $y = 2(5)^x$

b. $y = 7(0.3)^x$

c. $y = 100(1.08)^x$

3. Identify a and b for each exponential function $y = a(b)^x$. (Identify a and b)

a. $y = 6(3)^x$

b. $y = 0.5(\frac{1}{4})^x$

4. For $f(x) = 3(2)^x$, complete the table for $x = -1$, 0, 1, 2, 3. (Table)

5. For $y = 5(2)^x$, state the domain, the range, the y-intercept, and the equation of the asymptote. (Key Features)

6. Determine whether the data shows exponential behaviour. Explain. (Exponential Behaviour)

a. x: 0, 1, 2, 3 y: 4, 12, 36, 108

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7. Determine whether the data shows exponential behaviour. Explain. (Exponential Behaviour)

a. x: 0, 1, 2, 3   y: 5, 9, 13, 17

8. Each time you fold a piece of paper in half, the number of layers doubles. Start with 1 layer. (Apply)

a. Write a function for the number of layers after x folds.

b. How many layers after 7 folds?

9. A car worth $24,000 loses 15% of its value each year. (Apply)

a. Write an exponential function for its value after x years.

b. Find its value after 3 years.

10. A colony of 200 bacteria triples every hour. (Apply)

a. Write a function for the population after x hours.

b. How many bacteria after 4 hours?

11. Error Analysis – Gus says $y = 3(0.5)^x$ is exponential growth because 3 is positive. Identify his mistake. (Error Analysis)

12. Explain what an asymptote is, and why the graph of $y = a(b)^x$ never touches the x-axis when $a \neq 0$. (Explain)